

Review BIS604

September 6, 2025

Data Redundancy

- Unnecessarily storing the same data at different places
 - Islands of information (i.e., scattered data locations)
 - Increases the probability of having different versions of the same data
- Possible results of uncontrolled data redundancy
 - Poor data security
 - Data inconsistency
 - Data-entry errors
 - Data integrity problems

Data Anomalies

- Develop when not all of the required changes in the redundant data are made successfully
 - Update anomalies
 - Insertion anomalies
 - Deletion anomalies

Insertion Anomalies

EMPLOYEE-DEPARTMENT

<u>Employee_id</u>	Name	Dnumber	Dname	Manager
110	Jim	2	Finance	Kim
111	Kim	5	Account	Jim
112	Tim	5	Account	Jim
NULL	NULL	10	Sales	James

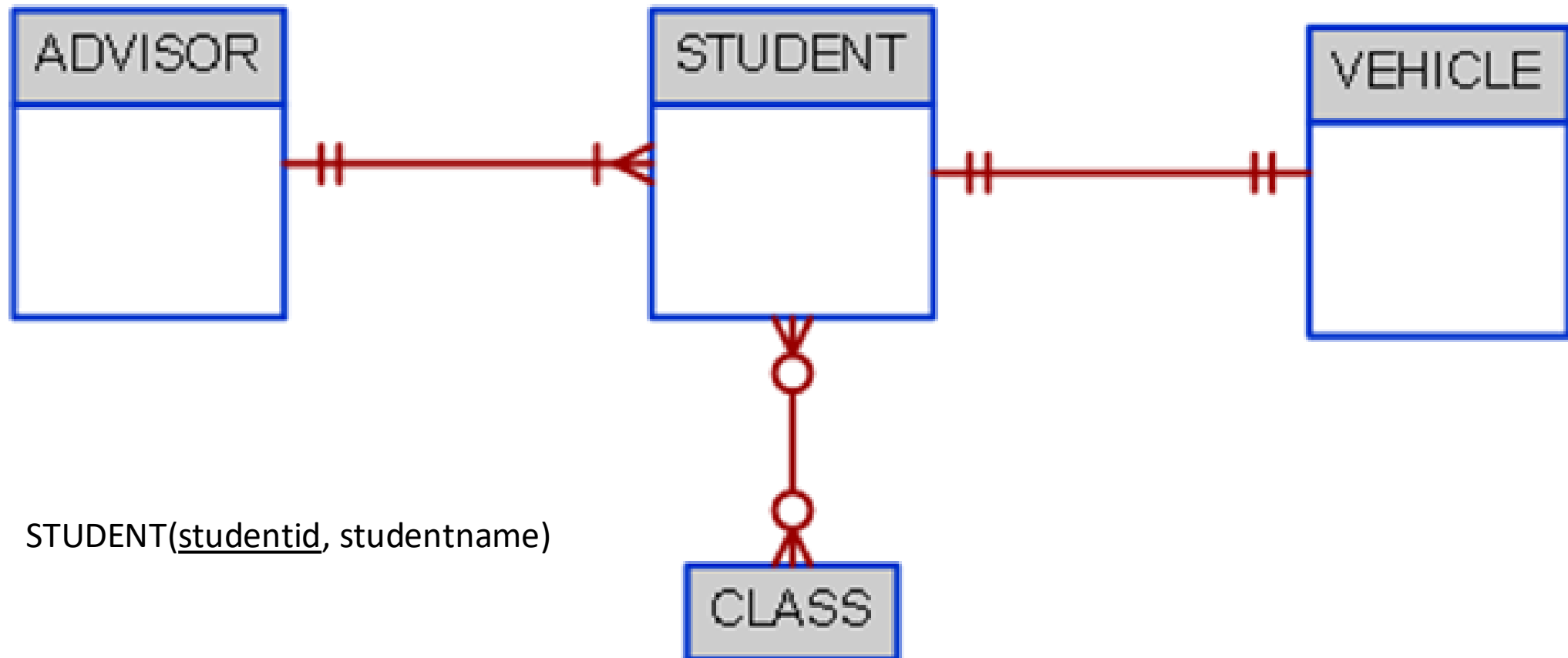
From the table, what should be recommended to make the data more suitable. Discuss. Also show the new design of data model.

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CODE	JOB_CHG_HOUR	PROJ_HOURS	EMP_PHONE
1	Hurricane	101	John D. Newson	EE	85.00	13.3	653-234-3245
1	Hurricane	105	David F. Schwann	CT	60.00	16.2	653-234-1123
1	Hurricane	110	Anne R. Ramoras	CT	60.00	14.3	615-233-5568
2	Coast	101	John D. Newson	EE	85.00	19.8	653-234-3254
2	Coast	108	June H. Sattlemeir	EE	85.00	17.5	905-554-7812
3	Satellite	110	Anne R. Ramoras	CT	62.00	11.6	615-233-5568
3	Satellite	105	David F. Schwann	CT	26.00	23.4	653-234-1123
3	Satelite	123	Mary D. Chen	EE	85.00	19.1	615-233-5432
3	Satellite	112	Allecia R. Smith	EE	85.00	20.7	615-678-6879

Translating Business Rules into Data Model Components

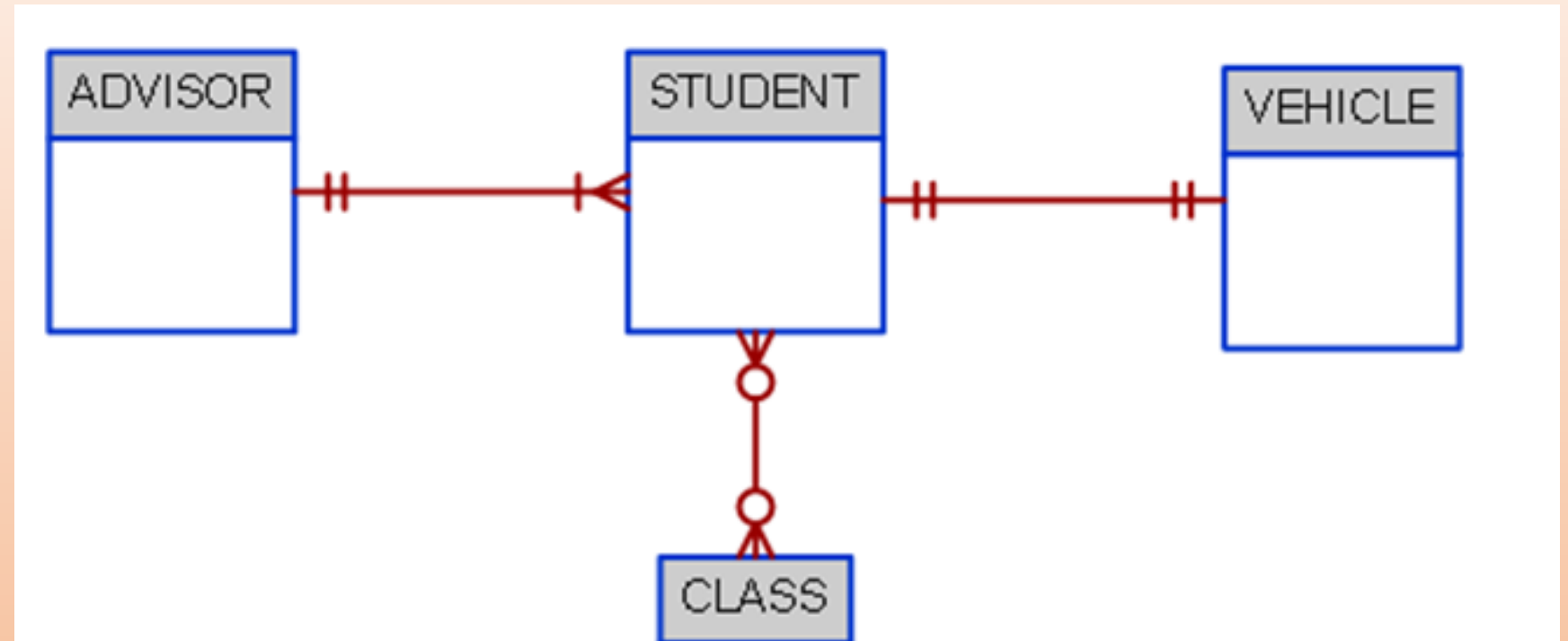
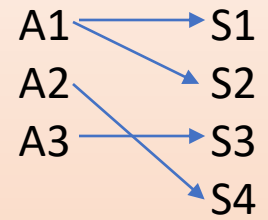
- Business rules set the stage for the proper identification of entities, attributes, relationships, and constraints
 - Nouns translate into entities
 - Verbs translate into relationships among entities
- Relationships are bidirectional
 - Questions to identify the relationship type
 - How many instances of B are related to one instance of A?
 - How many instances of A are related to one instance of B?

Write the business rules based on the diagram.
Also build a bridge entity to solve the M:N
relationship

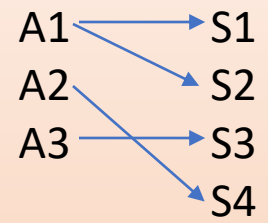


Consider the following example of data related together among the ADVISOR and STUDENT entities.
Show all the data of the remaining relationship diagram.

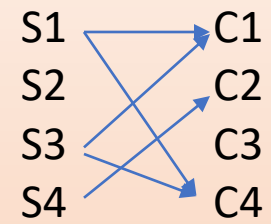
ADVISOR – STUDENT



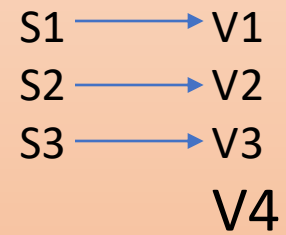
ADVISOR – STUDENT



STUDENT – CLASS



STUDENT – VEHICLE



COURSE – CLASS

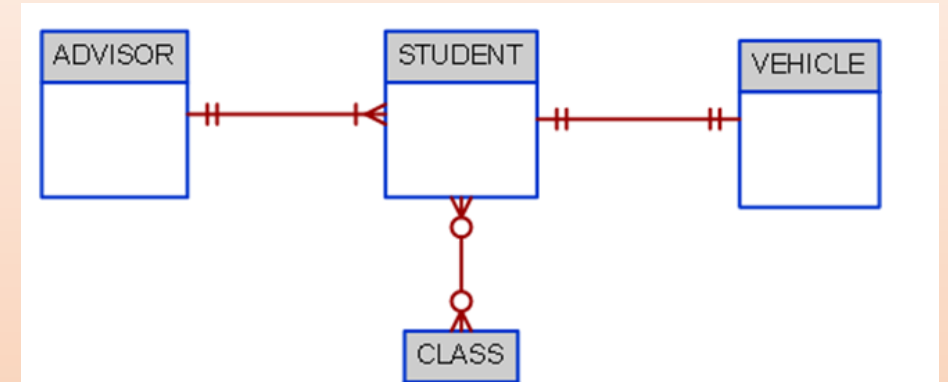
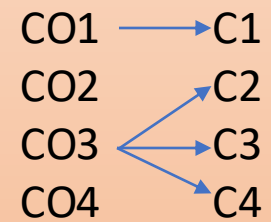
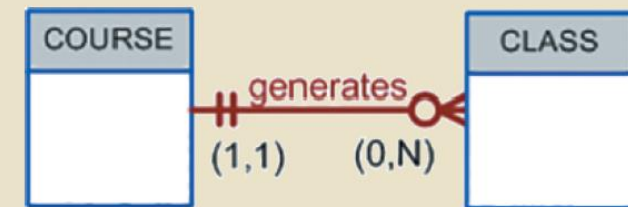


FIGURE 4.13 CLASS IS OPTIONAL TO COURSE



Attributes

- Characteristics of entities
 - Required attribute: must have a value and cannot be left empty
 - Optional attribute: does not require a value and can be left empty
 - Domain: set of possible values for a given attribute
 - Identifier: one or more attributes that uniquely identify each entity instance
 - Composite identifier: primary key composed of more than one attribute
 - Composite attribute: attribute that can be subdivided to yield additional attributes
 - Simple attribute: attribute that cannot be subdivided
 - Single-valued attribute: attribute that has only a single value
 - Multivalued attributes: attributes that have many values

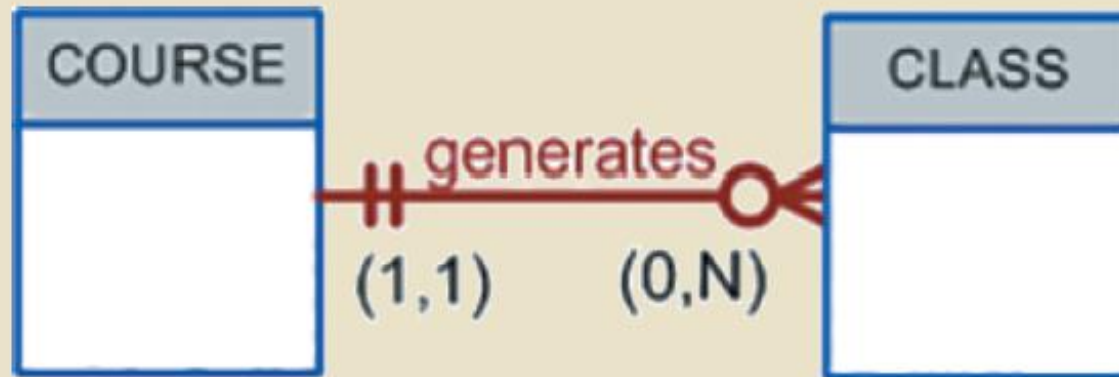
Relationships, Connectivity, and Cardinality

- Association between entities that always operate in both directions
 - Participants: entities that participate in a relationship
- Connectivity: describes the relationship classification
 - Include 1:1, 1:M, and M:N
- Cardinality: expresses the minimum and maximum number of entity occurrences associated with one occurrence of related entity
 - In the ERD, cardinality is indicated by placing the appropriate numbers beside the entities, using the format (x, y)

Relationship Strength

- Weak (non-identifying) relationship
 - Primary key of the related entity does not contain a primary key component of the parent entity
- Strong (identifying) relationships
 - Primary key of the related entity contains a primary key component of the parent entity

FIGURE 4.13 CLASS IS OPTIONAL TO COURSE

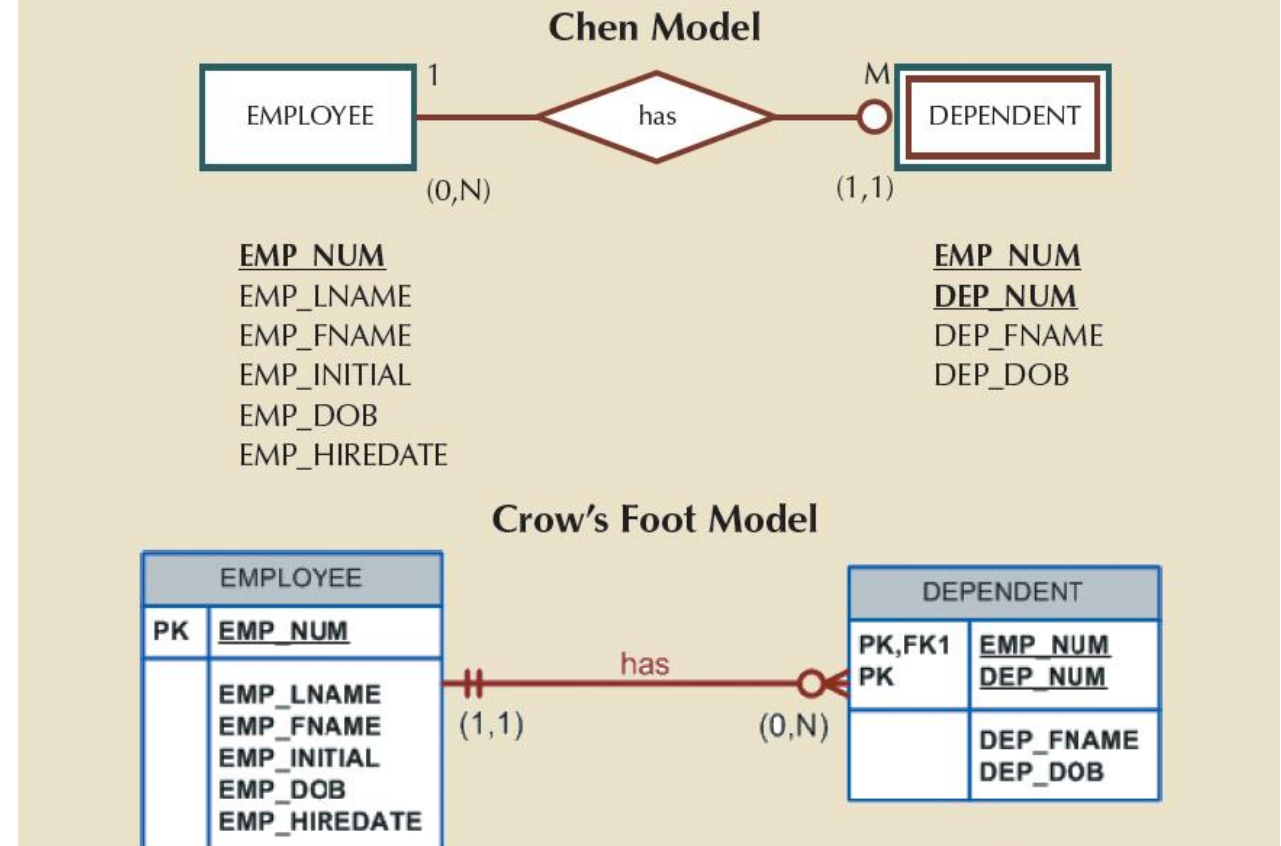


Weak Entities

- Conditions of a weak entity
 - Existence-dependent
 - Has a primary key that is partially or totally derived from parent entity in the relationship
- Database designer determines whether an entity is weak
 - Based on business rules

Existence Dependence	
• Existence dependence	
• Entity exists in the database only when it is associated with another related entity occurrence	
• Existence independence	
• Entity exists apart from <u>all</u> of its related entities	
• Referred to as a strong entity or regular entity	

FIGURE 4.10 A WEAK ENTITY IN AN ERD



Dependencies

- Determination
 - State in which knowing the value of one attribute makes it possible to determine the value of another
 - Establishes the role of a key
 - Based on the relationships among the attributes
- Functional dependence: value of one or more attributes determines the value of one or more other attributes
 - Determinant: attribute whose value determines another
 - Dependent: attribute whose value is determined by the other attribute
- Full functional dependence: entire collection of attributes in the determinant is necessary for the relationship

EMPID → Name, Salary, DeptID

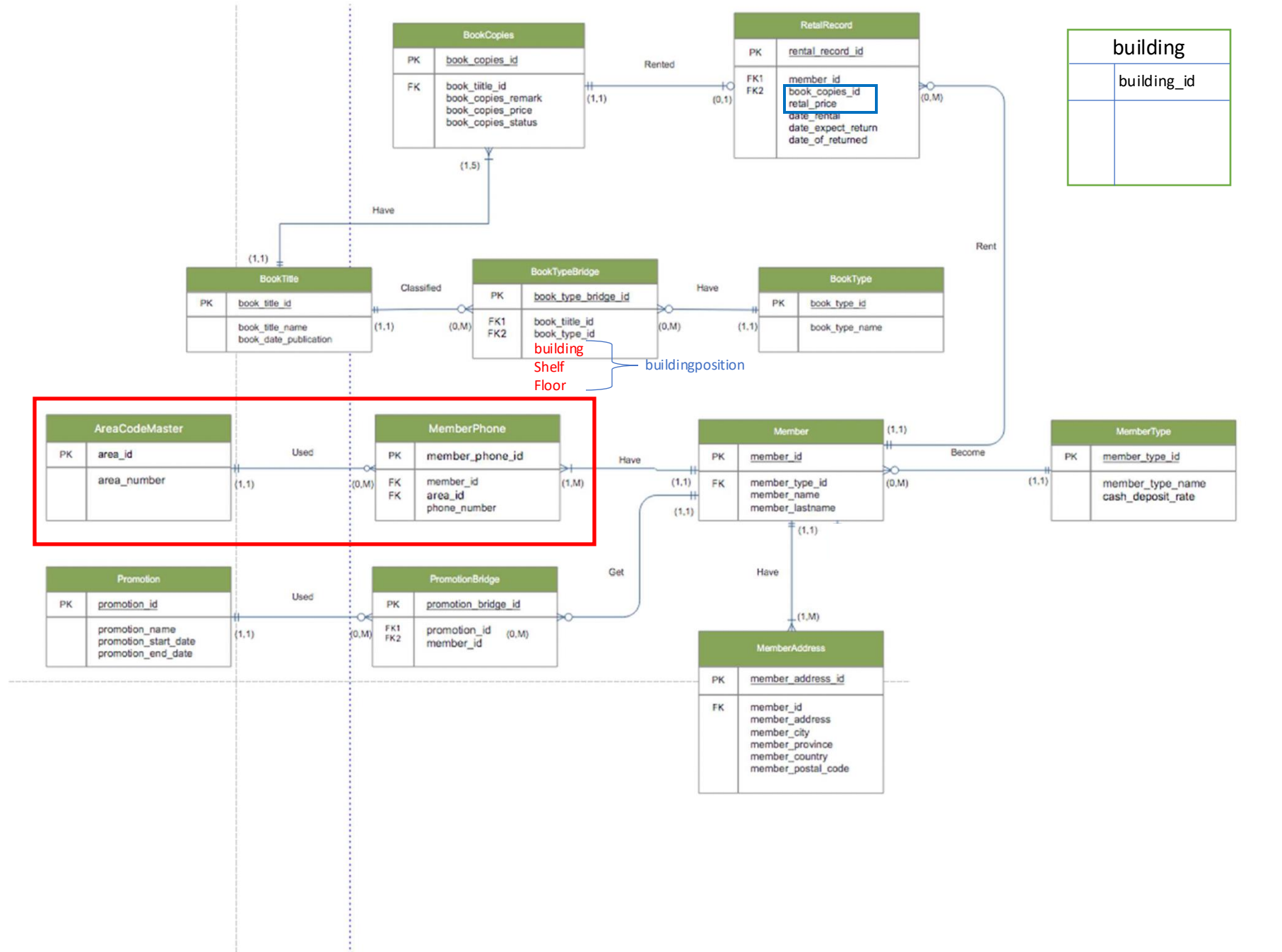
EMPID → Name, Salary, DeptID
EMPID, SkillID → DatCompleted

Types of Keys (1 of 3)

- Several different types of keys are used in the relational model
 - Composite key: key that is composed of more than one attribute
 - Key attribute: attribute that is a part of a key
 - Superkey: key that can uniquely identify any row in the table
 - Candidate key: minimal superkey
 - Entity integrity: condition in which each row in the table has its own unique identity
 - All of the values in the primary key must be unique
 - No key attribute in the primary key can contain a null
 - Null: absence of any data value
 - Unknown attribute value, known but missing attribute value, or inapplicable condition
 - Referential integrity: every reference to an entity instance by another entity instance is valid
 - Foreign key: primary key of one table that has been placed into another table to create a common attribute
 - Secondary key: key used strictly for data retrieval purposes

1	2	3
STU_NAME	CLASS_NAME	ROOM_NAME
A	Circle	I
A	Circle	II
B	Circle	III
B	Square	III
C	Circle	II

1,2 $\rightarrow$ 3



The local city youth league needs a database system to help track children that sign up to play soccer. Data needs to be kept on each team and the children that will be playing on each team and their parents. Also, data needs to be kept on the coaches for each team. Each team will have team symbol or team colors (some team has only one or more than one colors). Draw the data model described below. Resolve the M:M, the multivalued attribute, and other issues, if any.

The questions/report that must be answer/report from this database include:

1. Who are the players that has the same parents
2. List all the name of coach of each team
3. Who are the players in Team A. (assume your data has team A)
4. What is the symbol of team A. (assume your data has team A)
5. Which team has the same team color. (the same color for only one color or all colors must be listed as result)

Answer the following questions:

1. Define the business rules for identifying each connectivity and participatory.
2. Show the completed ERD segment

The local city youth league needs a database system to help track children that sign up to play soccer. Data needs to be kept on each team and the children that will be playing on each team and their parents. Also, data needs to be kept on the coaches for each team. Draw the data model described below. Resolve the M:M, the multivalued attribute, and other issues, if any.

Entities required: Team, Player, Coach, and Parent.

Attributes required:

Team: Team ID number, Team name, and Team colors.

Player: Player ID number, Player first name, Player last name, and Player age.

Coach: Coach ID number, Coach first name, Coach last name, and Coach home phone number.

Parent: Parent ID number, Parent last name, Parent first name, Home phone number, and Home Address (Street, City, State, and ZIP Code).

Connectivities and participations are defined as follows:

A Team may or may not have a Player.

A Player must have a Team.

A Team may have many Players.

A Player has only one Team.

A Team may or may not have a Coach.

A Coach must have a Team.

A Team may have many Coaches.

A Coach has only one Team.

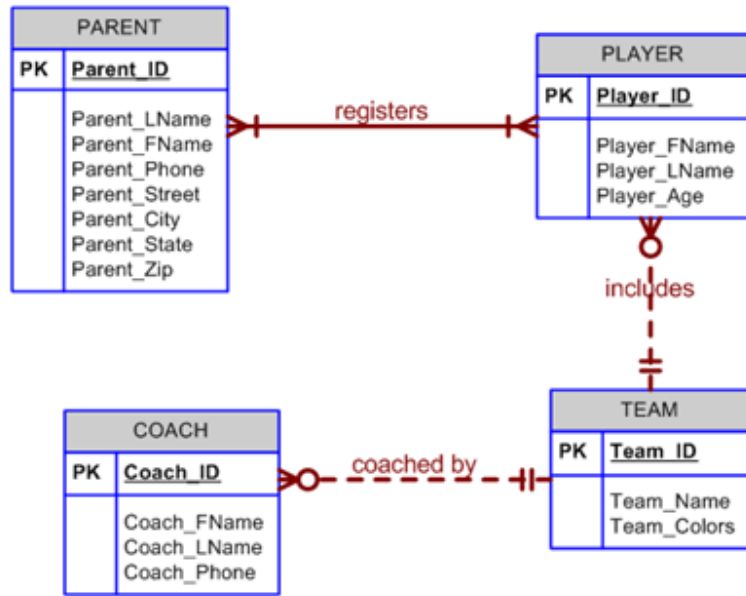
A Player must have a Parent.

A Parent must have a Player.

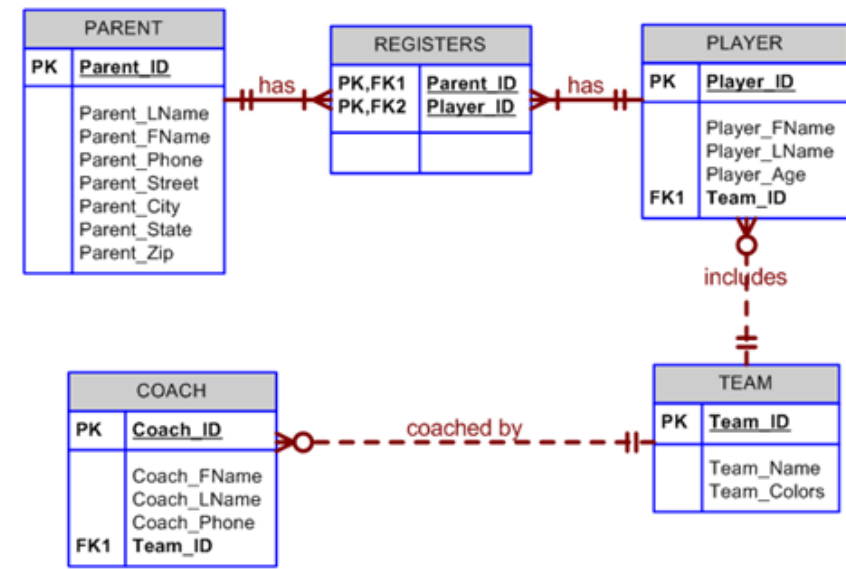
A Player may have many Parents.

A Parent may have many Players.

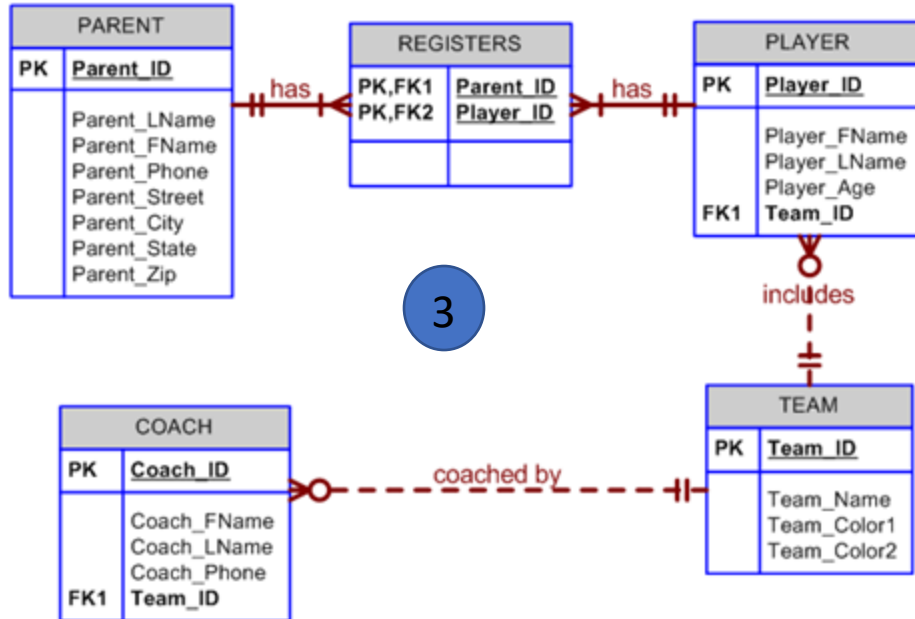
1



2



3



4

